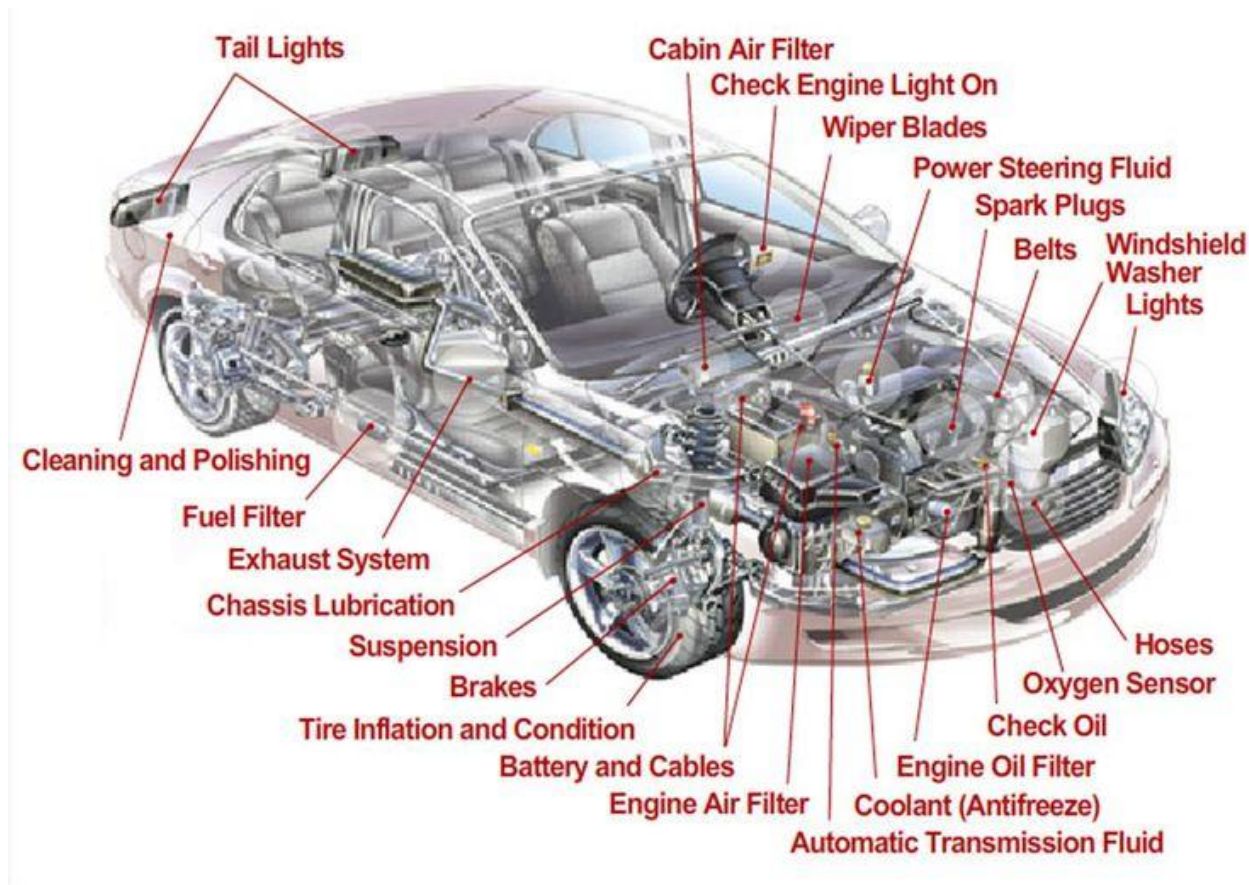
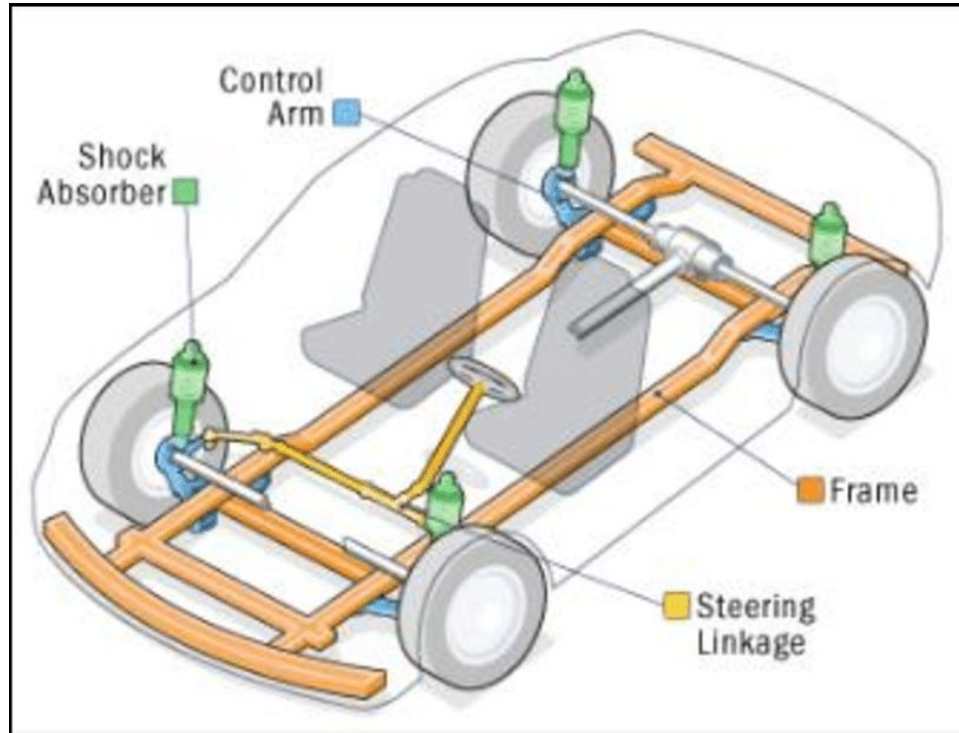


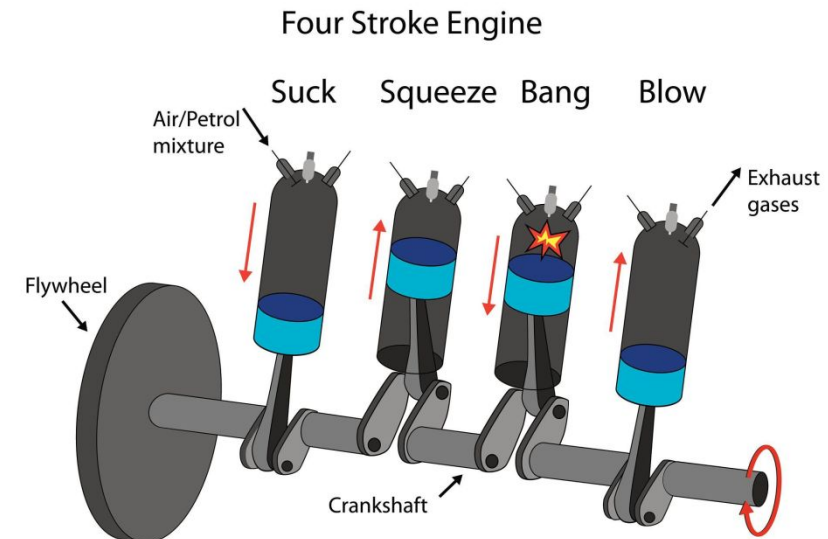
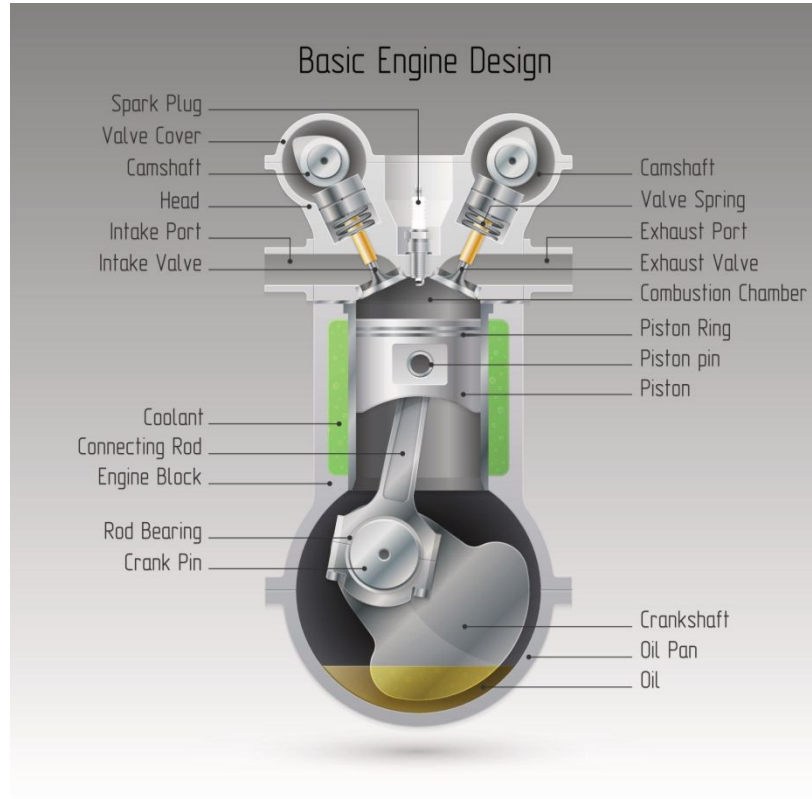
Mechanical Overview of Vehicle :



Mechanical Overview of Vehicle : Chassis



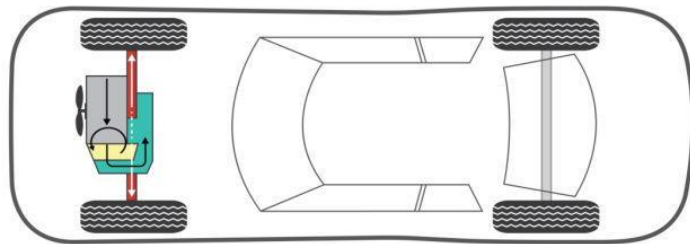
Mechanical Overview of Vehicle : Engine



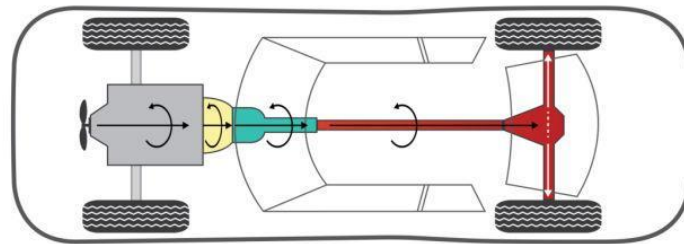
1. Suck - Induction Stroke - Air/Petrol mixture drawn in.
2. Squeeze - Compression Stroke - Valves close and piston moves up.
3. Bang - Power Stroke - Electric spark from spark plug ignites mixture.
4. Blow - Exhaust Stroke - Piston moves up pushing gases out.

Mechanical Overview of Vehicle : Transmission

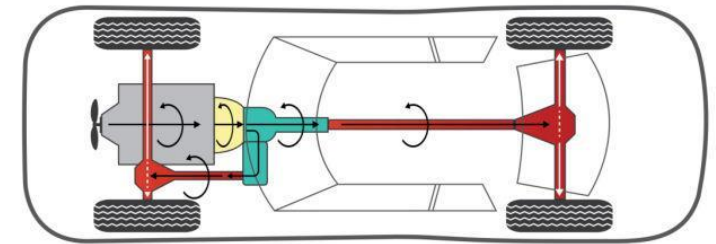
Transmission System - Layout



Front Wheel Drive (FWD) Layout



Rear Wheel Drive (RWD) Layout



Four Wheel Drive (4WD) Layout

● Engine ● Transmission
● Torque Converter ● Final Drive

Mechanical Overview of Vehicle : Braking System

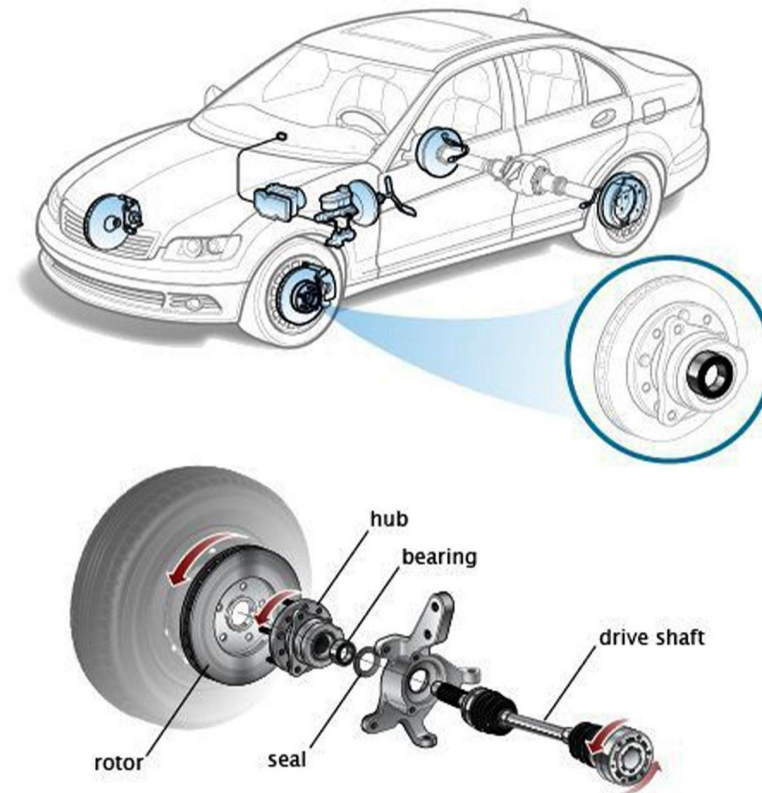
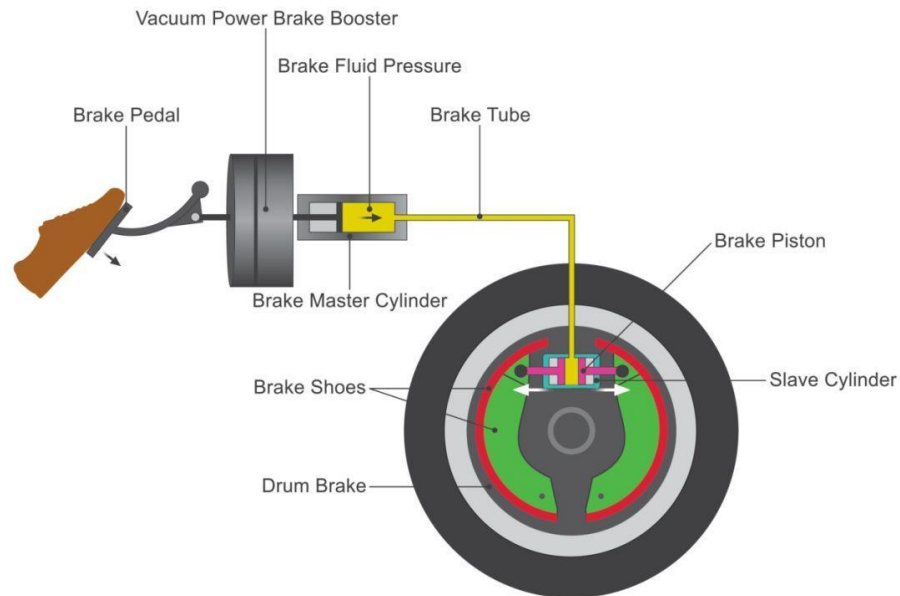


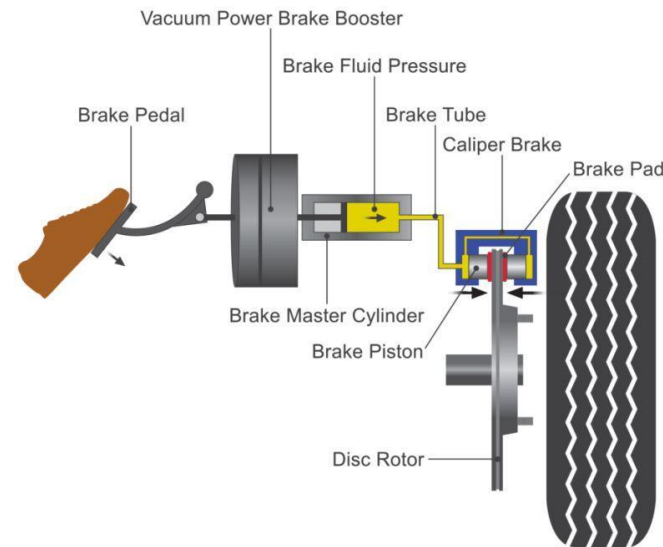
Image courtesy of ClearMechanic.com

Mechanical Overview of Vehicle :

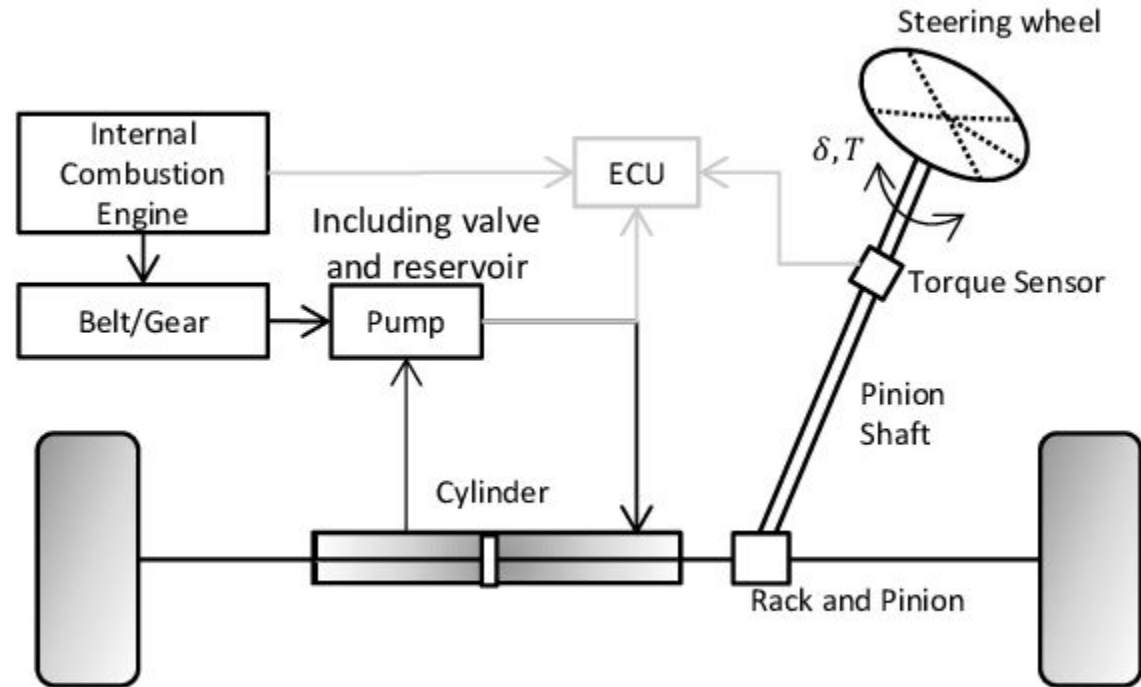
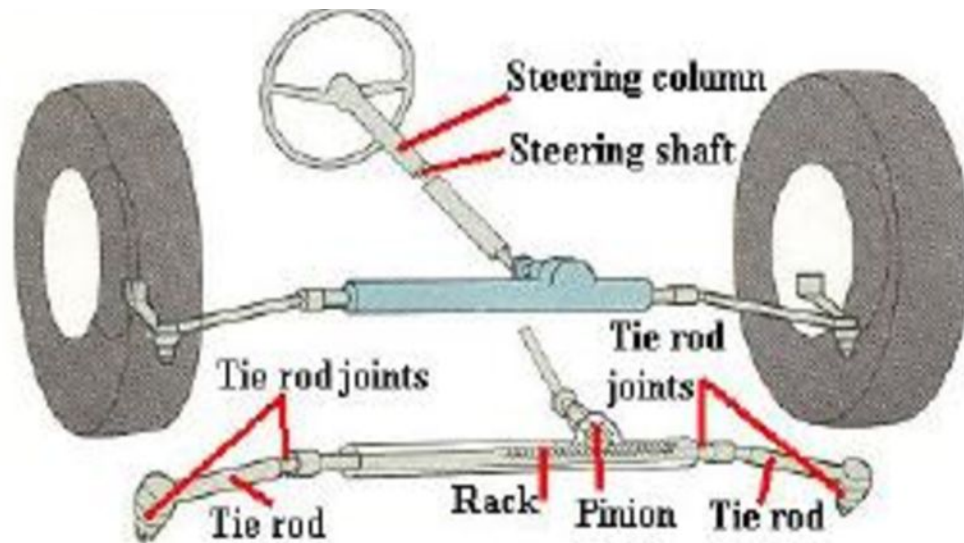
Hydraulic Drum Brake System



Hydraulic Disc Brake System

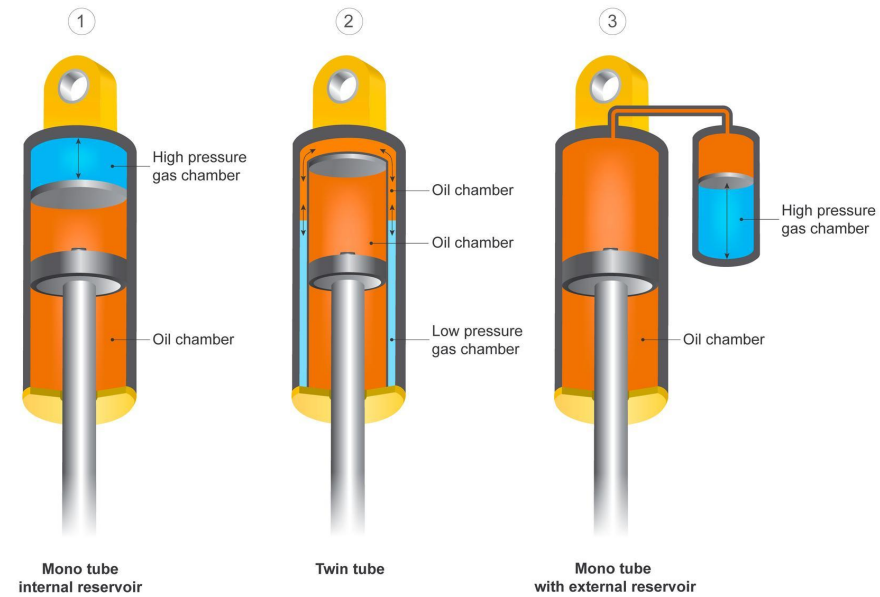


Mechanical Overview of Vehicle : Steering System

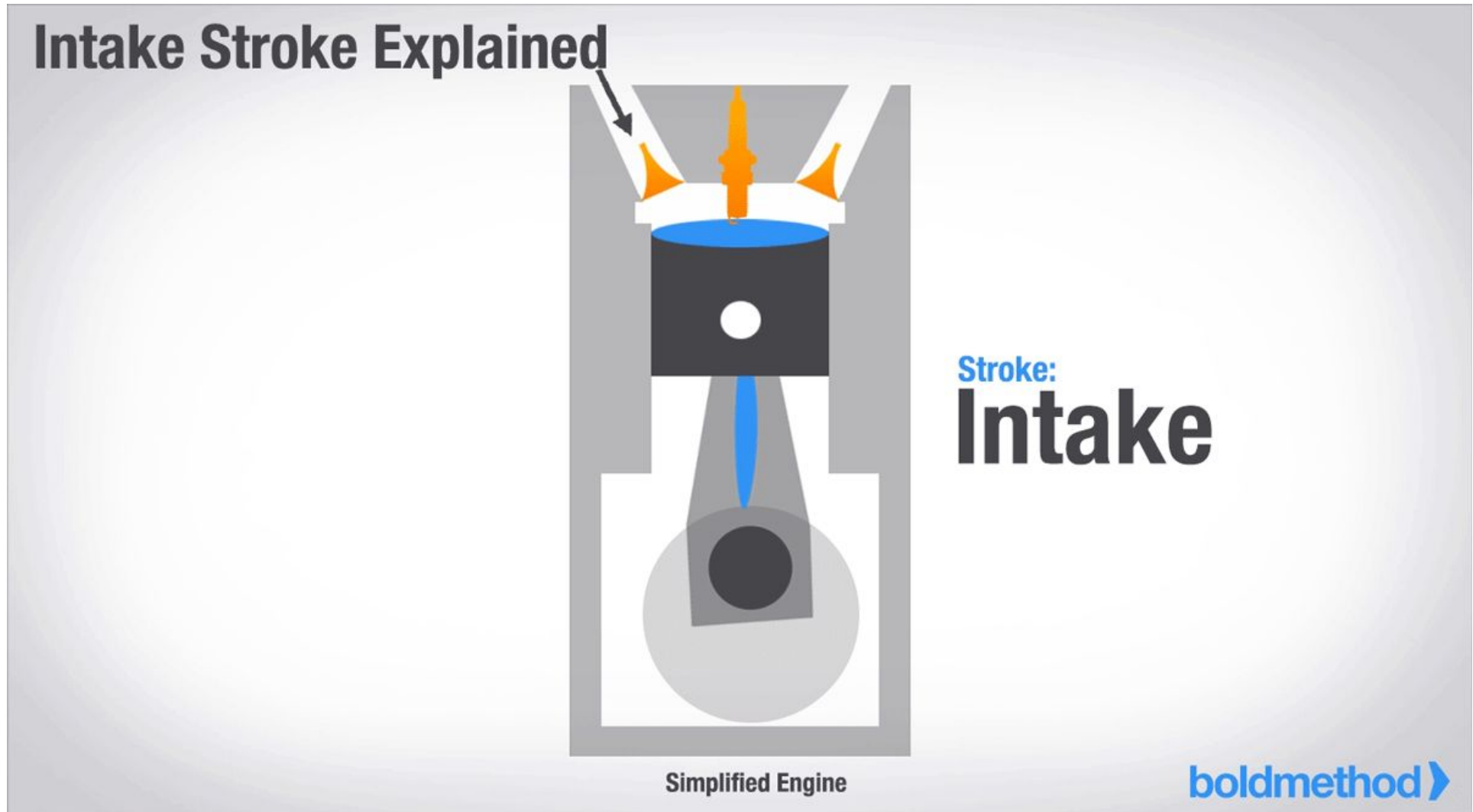


Mechanical Overview of Vehicle :

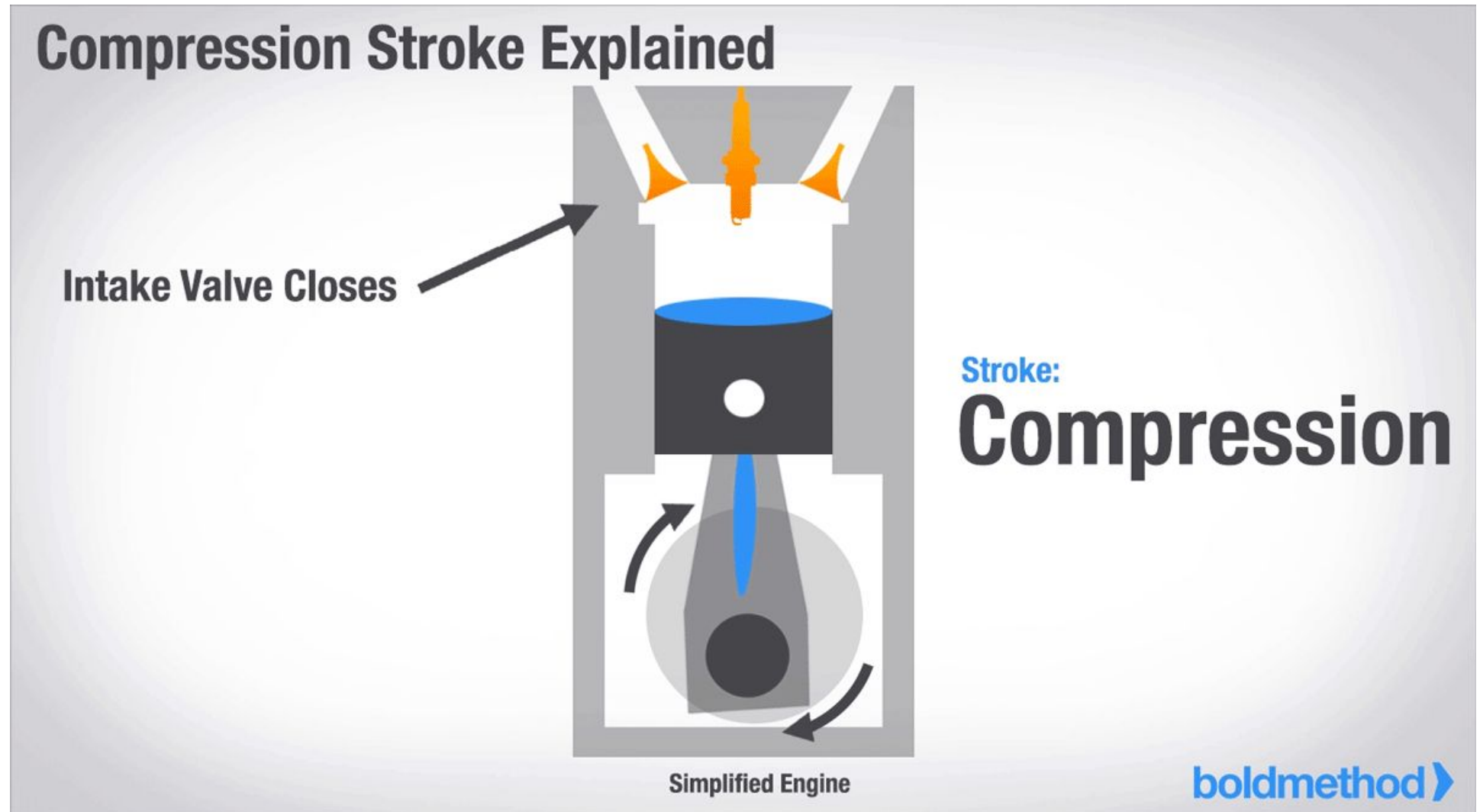
Types of Shock Absorber



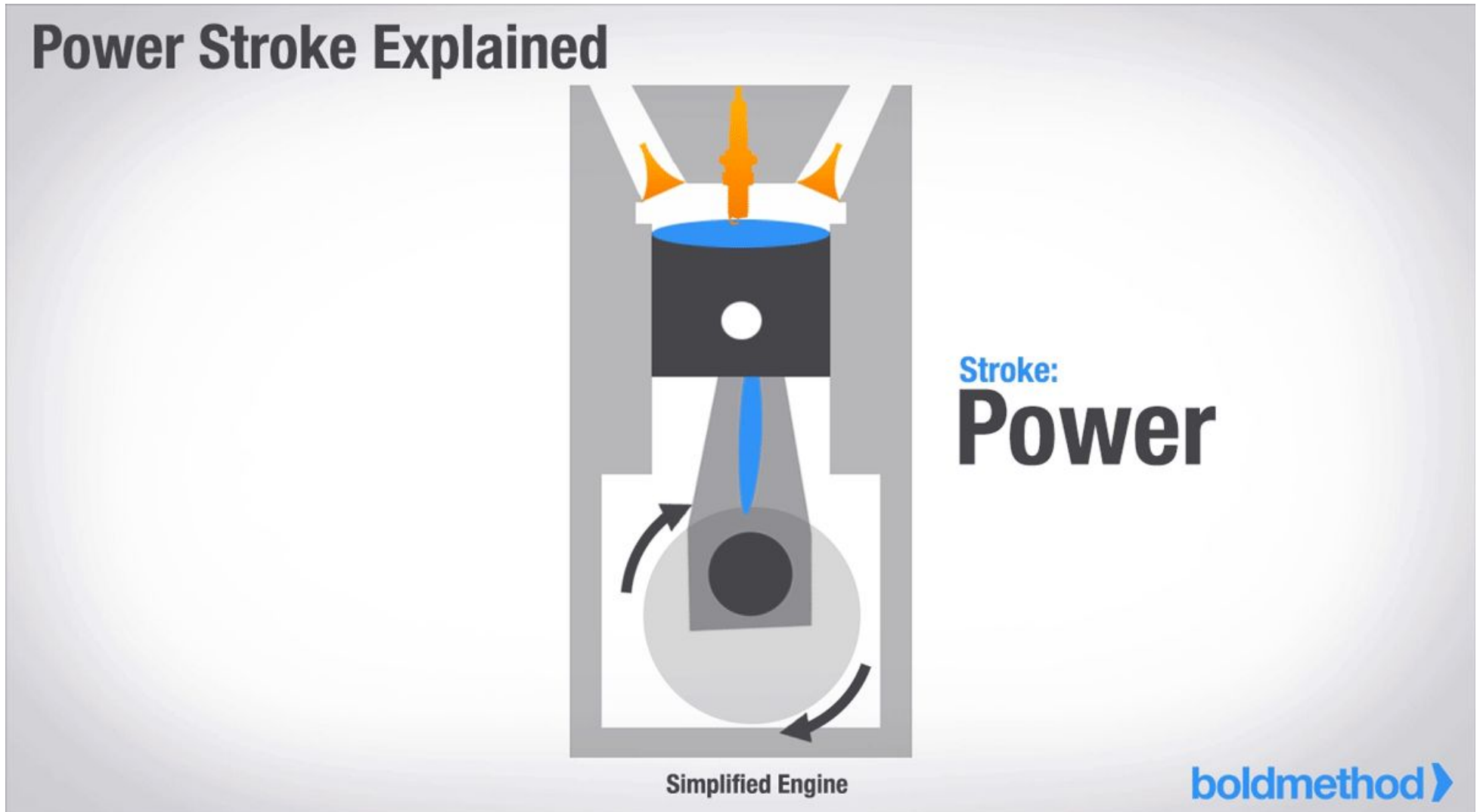
Four Stroke Engine :



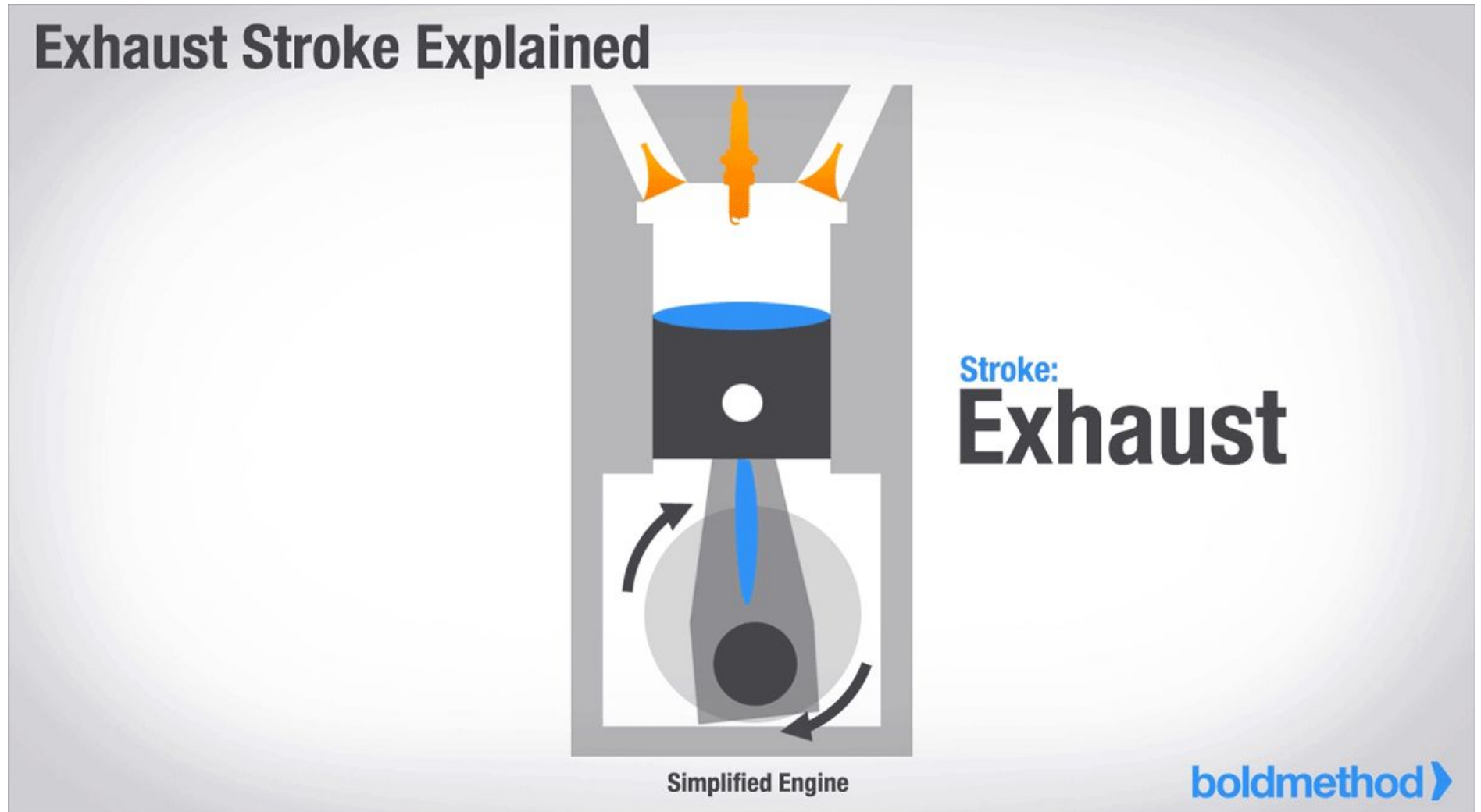
Four Stroke Engine :



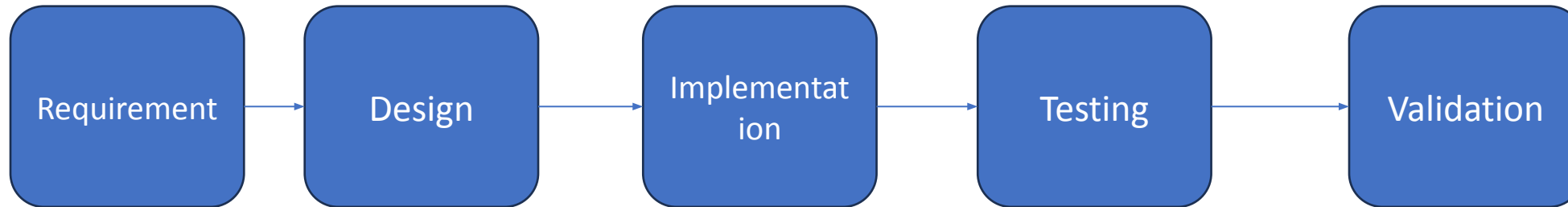
Four Stroke Engine :



Four Stroke Engine :



What is ECU Design Cycle?



This process must ensure:

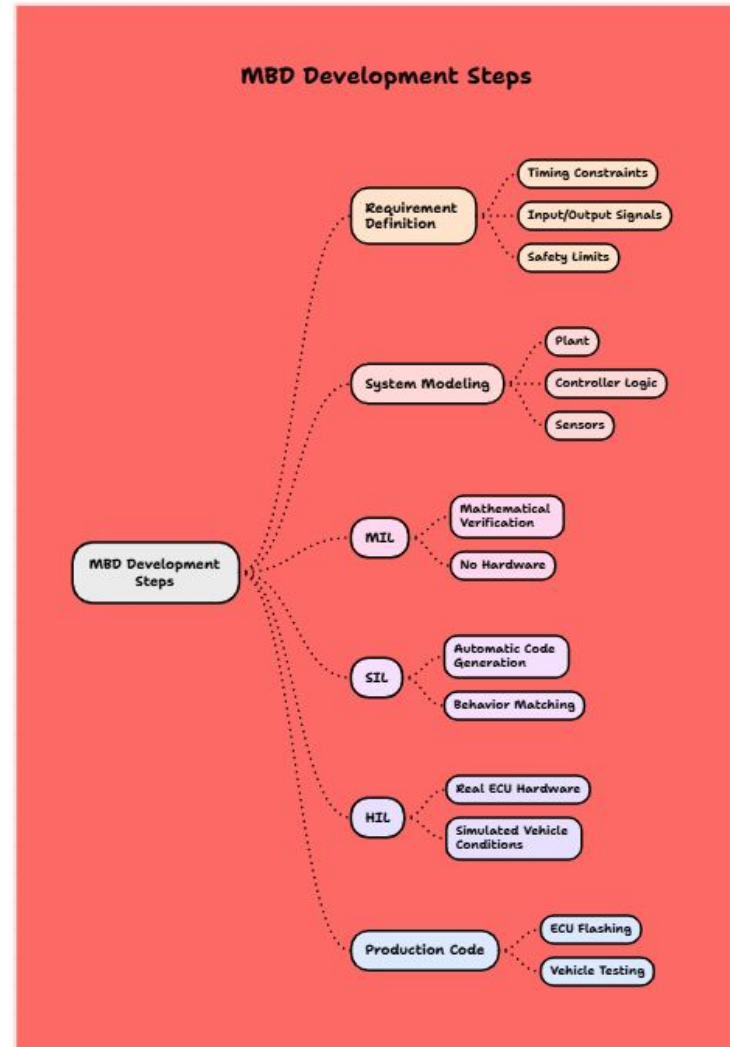
- Safety (ISO 26262)
- Real-time performance
- Reliability
- Traceability

ECU Design Cycle: Model Based Design (MBD)

Model Based Design means:
Designing the control system using
mathematical models before
writing embedded C code.

Used tools:

- MATLAB
- Simulink
- Stateflow



Why Automotive Uses MBD?

- Reduces development time.
- Early error detection.
- Automatic code generation.
- Easy requirement traceability.
- Supports safety standards.

ECU Design Cycle: V-Model

What is V-Model?

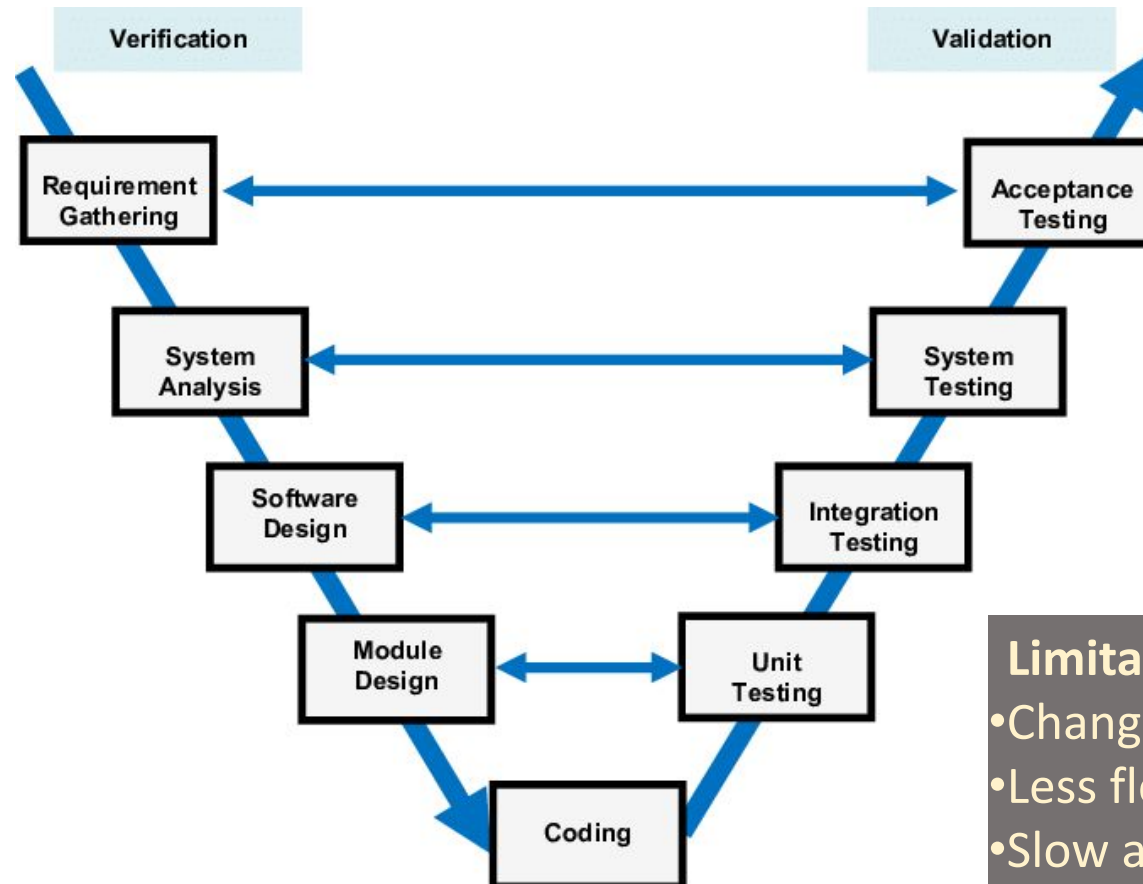
The V-model is a **structured, sequential development process**.

Left side = Design
Right side = Testing

It ensures every design stage has a corresponding test stage.

Why V-Model is Popular in Automotive?

- Strong documentation.
- Required for ISO 26262.
- Clear traceability.
- Suitable for safety-critical systems.



Limitation of V-Model

- Changes are expensive.
- Less flexible.
- Slow adaptation to new features.

What is Agile?

Agile is an **iterative and flexible development approach**.

Instead of one long cycle, work is divided into small cycles called **sprints**.

Highlights of Agile in Automotive?

- Software Defined Vehicles.
- Frequent updates.
- Connectivity features.
- Infotainment & ADAS updates.

Agile Steps

- Define small feature.
- Develop quickly.
- Test immediately.
- Improve in next sprint.

Cycle duration:

- 2–4 weeks typically.

Challenge of Agile in Automotive

- Safety documentation is heavy.
- Certification requirements.
- Hard to apply pure Agile for safety ECUs.

Aspect	V-Model	Agile
Structure	Fixed & sequential	Flexible & iterative
Documentation	Very strong	Moderate
Change handling	Difficult	Easy
Safety compliance	Excellent	Complex
Best suited for	ABS, Airbag	Infotainment, Connectivity

What is an Automotive-Grade ECU?

An automotive-grade ECU is:

A rugged, real-time embedded control unit designed to operate reliably under automotive conditions (-40°C to $+125^{\circ}\text{C}$, vibration, EMI, etc.).

It follows standards like:

- ISO 26262 (Functional Safety)
- AEC-Q100 (Automotive component qualification)
- AUTOSAR architecture (in many cases)

Powertrain ECU



These control vehicle movement.

Engine Control Unit (ECU / ECM)

- Controls fuel injection
- Controls spark timing
- Controls air-fuel ratio
- Monitors sensors (O₂, crankshaft, temperature)

Very critical ECU.

Transmission Control Module (TCM)

- Controls gear shifting
- Optimizes torque transfer
- Communicates with engine ECU

Motor Control Unit (MCU – EVs)

- Controls electric motor speed
 - Controls torque
 - Manages inverter switching
- Used in electric vehicles.

Battery Management System (BMS – EVs)

- Monitors battery voltage
- Monitors temperature
- Controls charging/discharging
- Ensures battery safety

Safety ECU



ABS ECU (Anti-lock Braking System)

Monitors wheel speed sensors
Prevents wheel lock
Controls brake pressure
Hard real-time ECU.

Airbag Control Module (ACM)

Detects crash using accelerometers
Deploys airbags within milliseconds
Very high safety level (ASIL-D)

Electronic Stability Control (ESC)

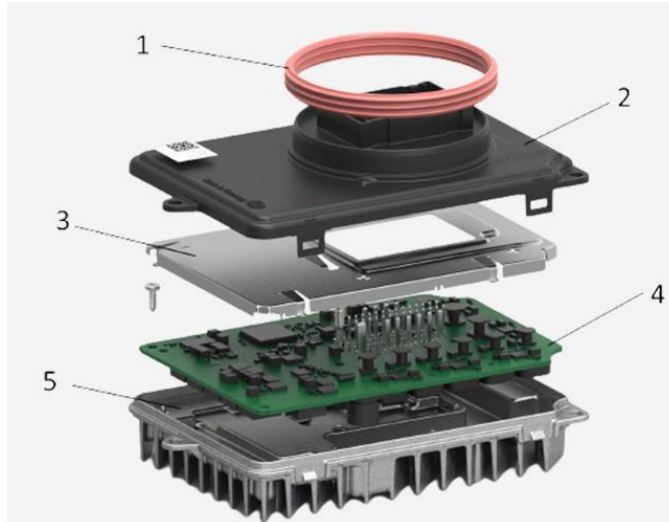
Prevents skidding
Controls individual wheel braking
Uses steering angle sensor + yaw sensor

ADAS ECU

Processes camera/radar data
Supports lane assist
Collision avoidance
Adaptive cruise control

Often high-performance processors.

Body & Comfort ECUs



Body Control Module (BCM)

- ✓ Controls:
 - Power windows
 - Central locking
 - Lights
 - Wipers
- ✓ Communicates via CAN/LIN

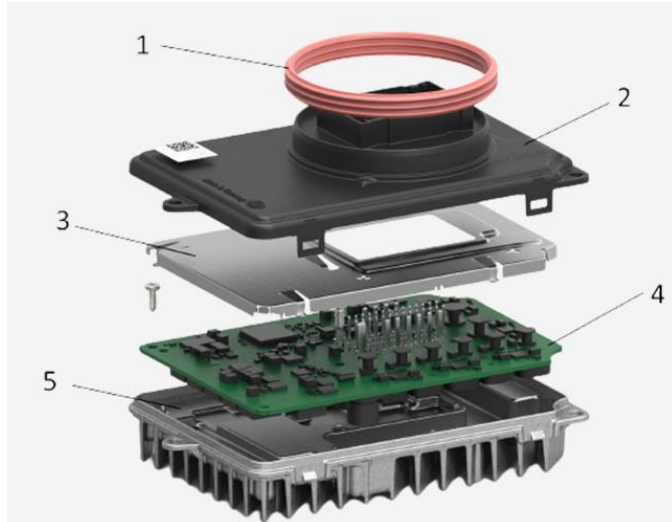
Lighting Control Module

- Headlamp control
- Adaptive headlights

Door Control Module

- Window motors
- Mirror adjustment
- Lock actuators

Infotainment & Connectivity ECUs



Infotainment Head Unit

- Touchscreen display
- Audio control
- Navigation
- Smartphone connectivity

Telematics Control Unit (TCU)

- GPS
- Cellular connectivity (4G/5G)
- Emergency call (eCall)

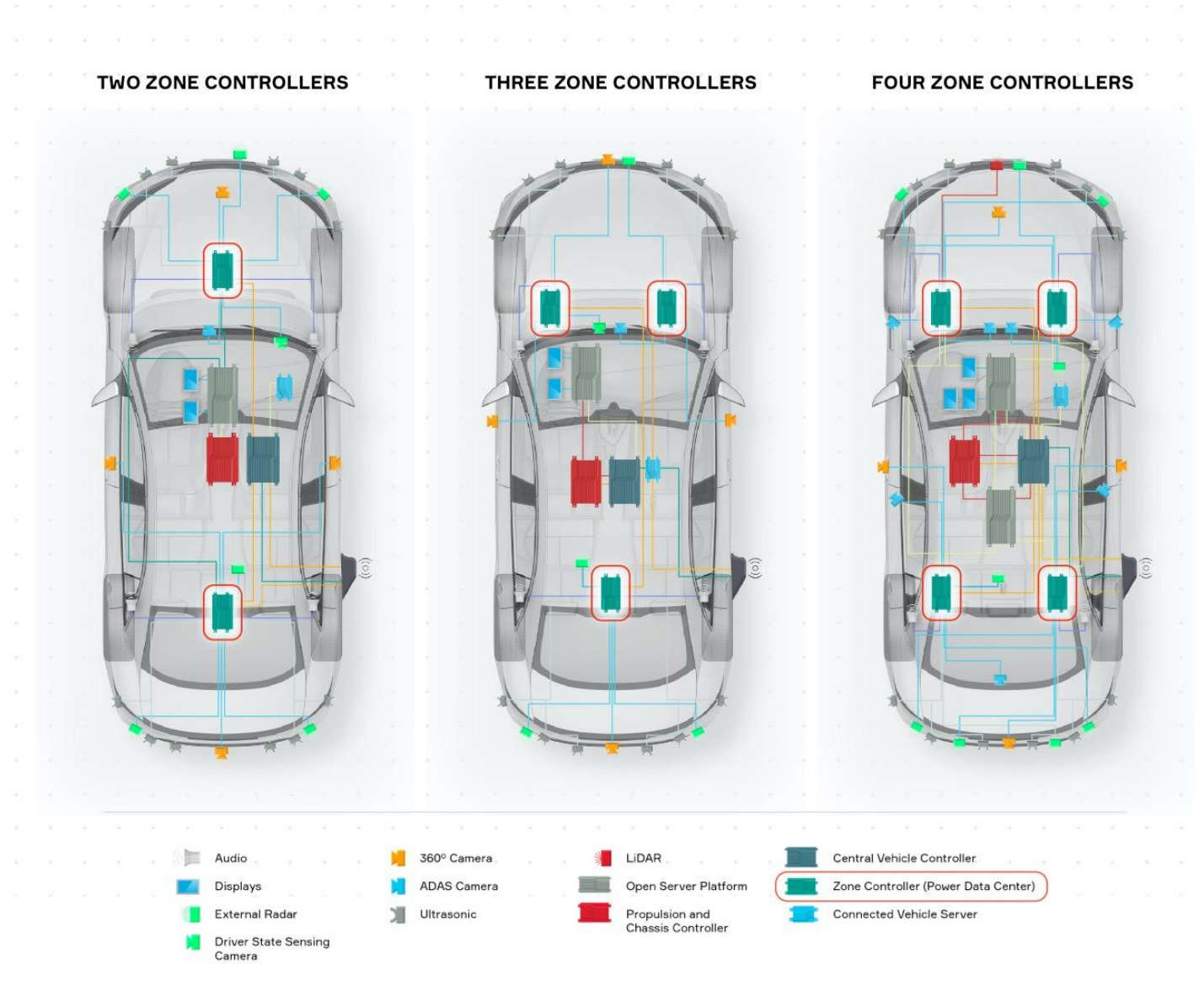
Gateway ECU

- Connects multiple networks:
 - CAN
 - LIN
 - FlexRay
 - Ethernet
- Manages secure communication

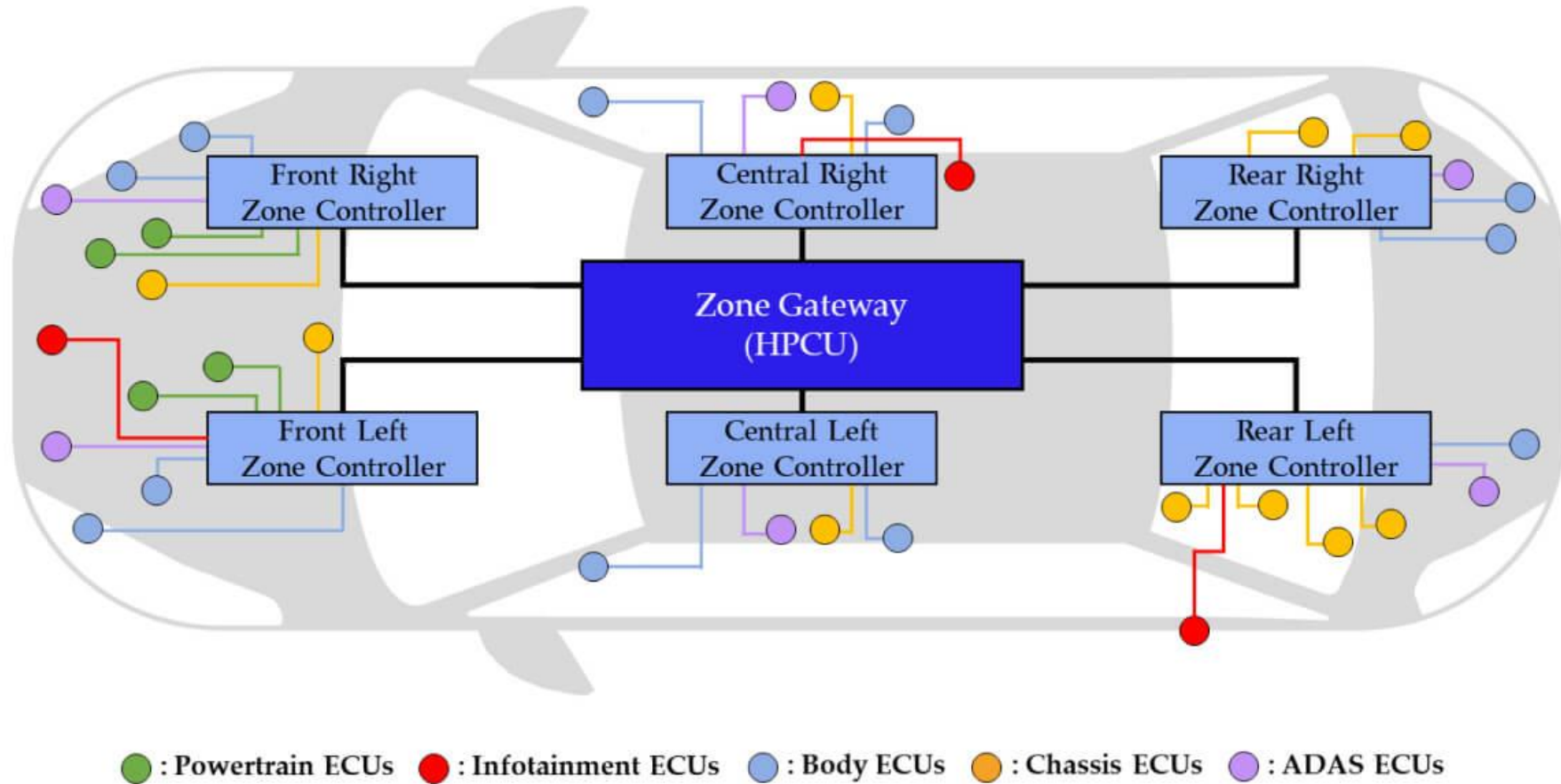
Digital Instrument Cluster ECU

- Displays speed
- RPM
- Warnings
- Driver information

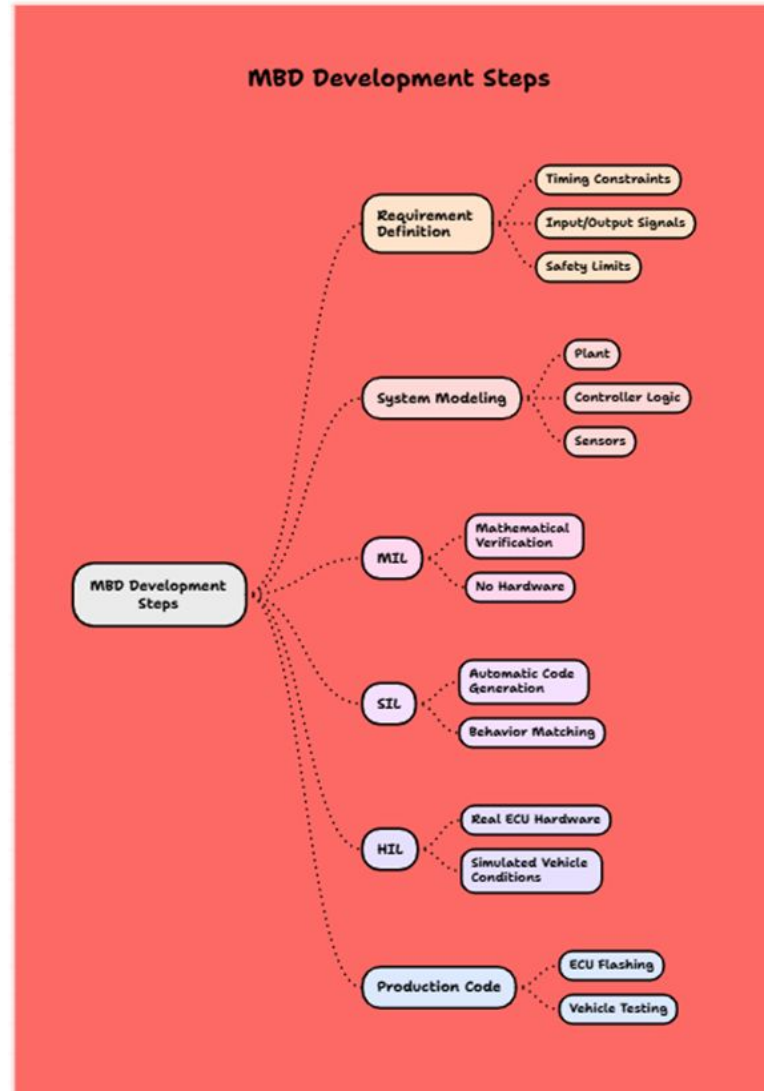
Modern ECUs



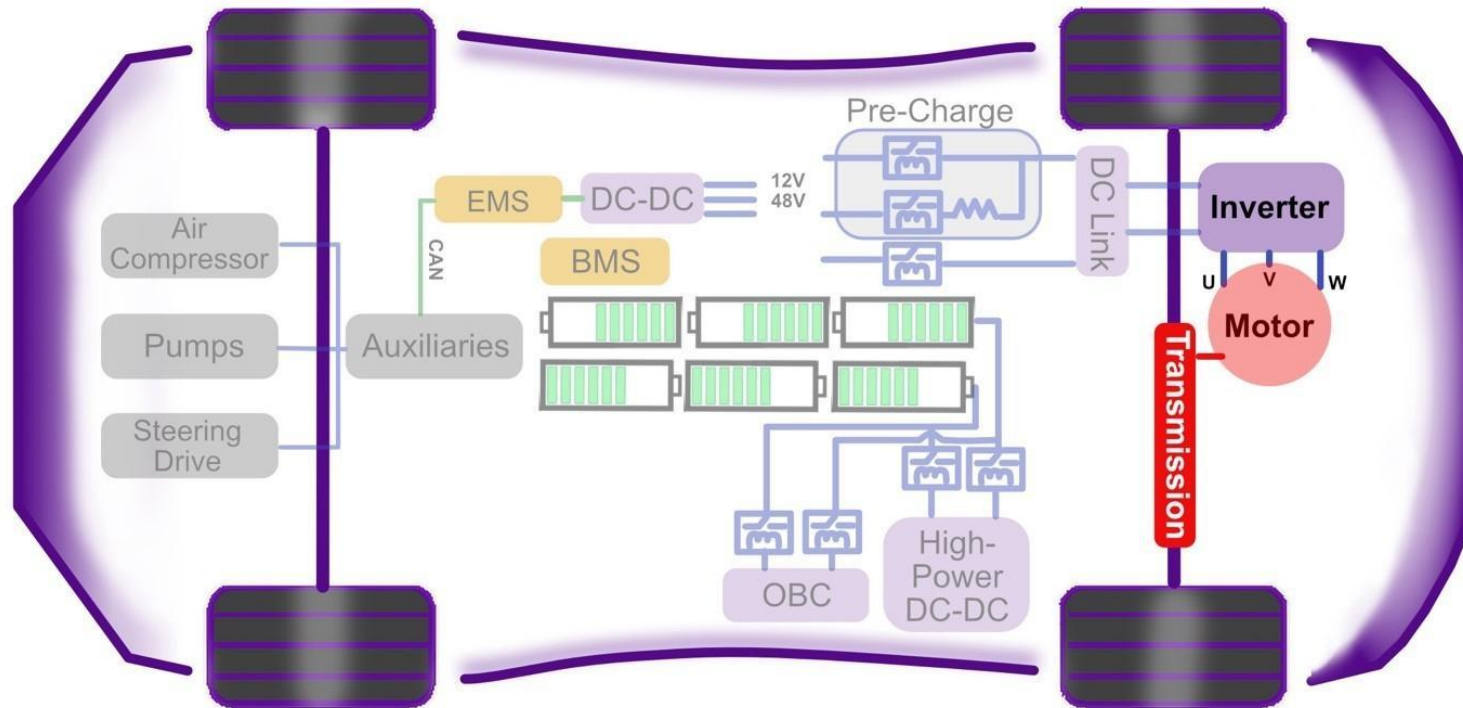
Modern ECUs



Modern ECUs



EVs



Instead of an engine, EV uses:

- Battery
- Electric motor
- Power electronics

EVs

Battery Pack

- Stores electrical energy.
- Usually Lithium-ion.
- Voltage range: 300V – 800V (modern EVs).

Electric Motor

- Converts electrical energy → mechanical rotation.
- Types:
 - PMSM (Permanent Magnet Synchronous Motor)
 - Induction Motor

Inverter

- Converts DC (battery) → AC (motor).
- Controls motor speed & torque.

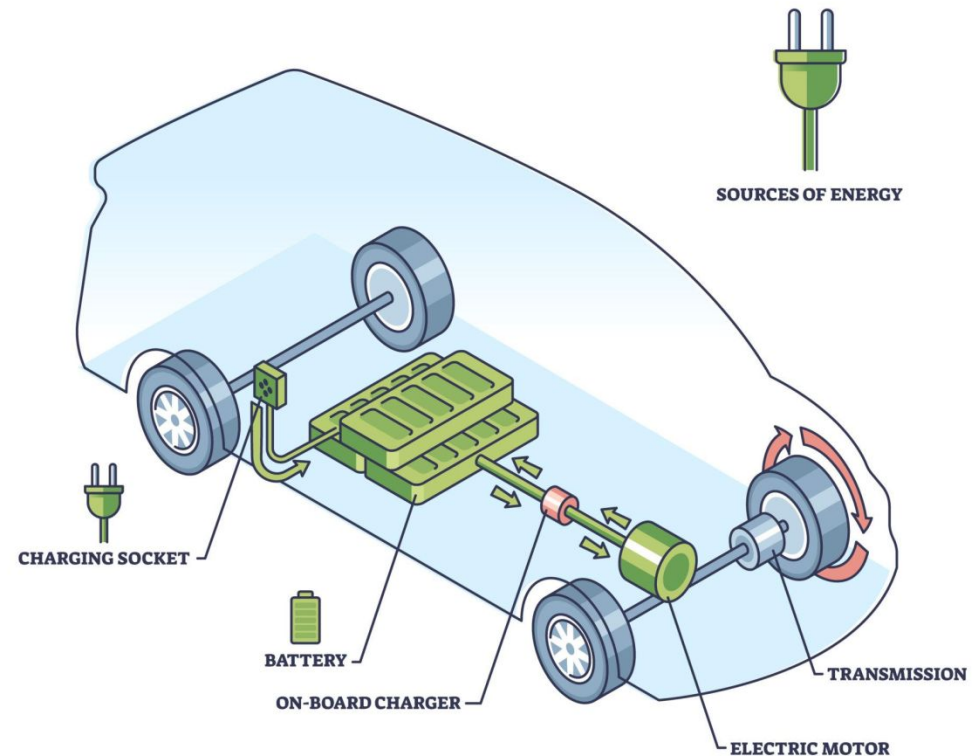
Battery Management System (BMS)

- Monitors voltage.
- Monitors temperature.
- Ensures battery safety.

On-Board Charger

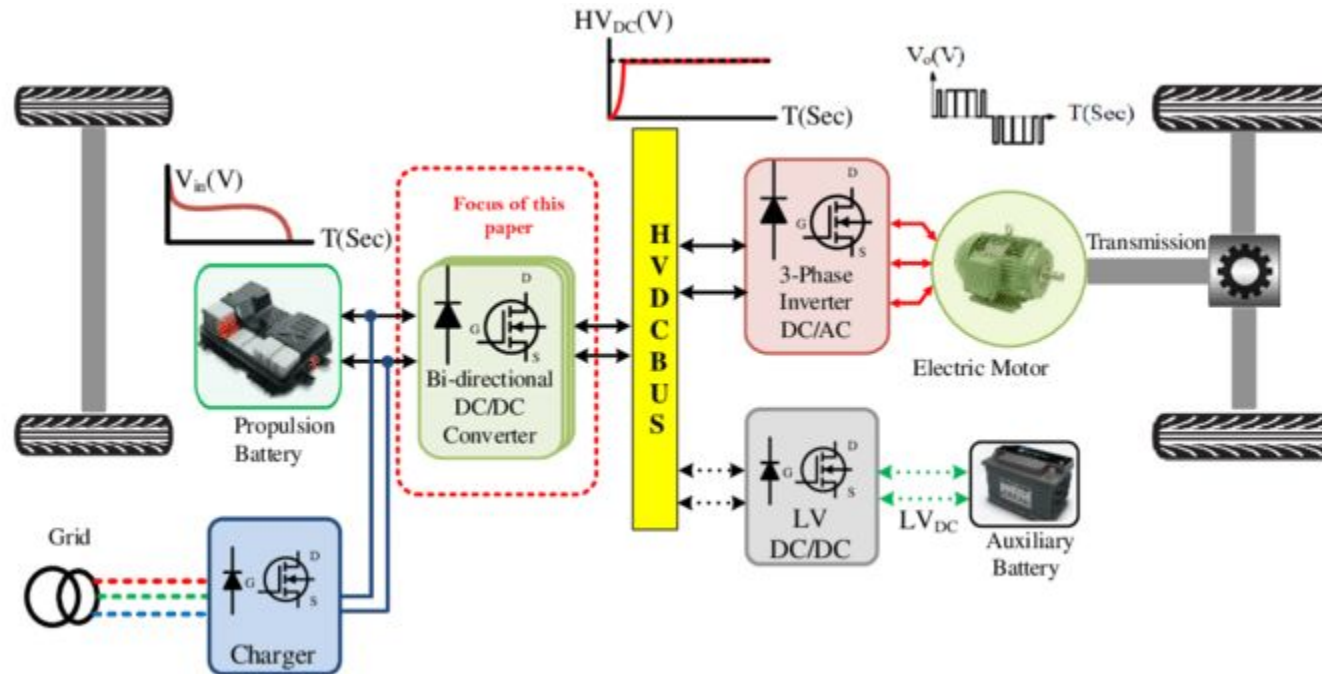
- Converts AC from grid → DC for battery charging.

BEV BATTERY ELECTRIC VEHICLE



EVs

EV Drive Train : The EV Drive Train is the system that converts electrical energy into vehicle motion. Unlike petrol cars, EVs have much fewer moving parts.



EVs

Battery Pack

- Stores electrical energy.
- High voltage: 300V–800V.
- Supplies DC power.

Inverter

- Converts DC (battery) → AC (motor).
- Controls:
 - Speed
 - Torque
 - Direction

Electric Motor

- Converts electrical energy → mechanical rotation.
- Common types:
 - PMSM (Permanent Magnet Synchronous Motor)
 - Induction Motor

Single-Speed Gearbox

- Reduces motor speed.
- Increases torque.
- EVs usually don't need multi-speed gearboxes.

Differential

- Splits torque between wheels.

Working of EV
Drive Train

Battery → Inverter → Motor → Gear Reduction → Wheels

EVs

Lithium-Ion (Most Common)

- High energy density.
- Lightweight.
- Long life.
- Used in most modern EVs.

Chemistry types:

- NMC (Nickel Manganese Cobalt)
- LFP (Lithium Iron Phosphate)

Solid-State Battery (Future)

- Solid electrolyte.
- Higher safety.
- Higher energy density.
- Still under development.

Lead-Acid (Old EVs)

- Low cost.
- Heavy.
- Short life.
- Mostly not used now.

EVs

TYPES OF ELECTRIC VEHICLES

